(iv) The rational number which is its own reciprocal is –1.

(v) The multiplicative inverse exists for all rational numbers.

3. Find the sum:

(i) $\frac{7}{5}+\frac{3}{5}$ (ii) $\frac{7}{5}+\frac{5}{7}$ (iii) $\frac{6}{5}+\left(\frac{-14}{15}\right)$ (iv) $-4\frac{2}{3}+7\frac{5}{12}$

4. Subtract : $\frac{-8}{44}$ from $\frac{-17}{11}$.

5. Evaluate : (i) $\frac{9}{132}\times\frac{-11}{3}$ (ii) $\frac{-7}{27}\times\frac{24}{-35}$

6. Divide : (i) $\frac{-21}{5}$ by $\frac{-7}{-10}$ (ii) $\frac{-3}{13}$ by –3 (iii) –2 by $\frac{-6}{15}$

7. Find $(a + b) \div (a - b)$ if

(i) $a=\frac{1}{2}, b=\frac{2}{3}$ (ii) $a=\frac{-3}{5}, b=\frac{2}{15}$

8. Simplify : $\frac{1}{2}+\left(\frac{3}{2}-\frac{2}{5}\right)\div\frac{3}{10}\times 3$ and show that it is a rational number between 11 and 12.

9. Simplify :

(i) $\left[\frac{11}{8}\times\left(\frac{-6}{33}\right)\right]+\left[\frac{1}{3}+\left(\frac{3}{5}\div\frac{9}{20}\right)\right]-\left[\frac{4}{7}\times\frac{-7}{5}\right]$ (ii) $\left[\frac{4}{3}\div\left(\frac{8}{-7}\right)\right]-\left[\frac{3}{4}\times\frac{4}{3}\right]+\left[\frac{4}{3}\times\left(\frac{-1}{4}\right)\right]$

10. A student had multiplied a number by $\frac{4}{3}$ instead of dividing it by $\frac{4}{3}$ and got 70 more than the correct answer. Find the number.

Objective Type Questions

11. The standard form of the sum $\frac{3}{4}+\frac{5}{6}+\left(\frac{-7}{12}\right)$ is _____ .

(A) 1 (B) $\frac{-1}{2}$ (C) $\frac{1}{12}$ (D) $\frac{1}{22}$

12. $\left(\frac{3}{4}-\frac{5}{8}\right)+\frac{1}{2}=$ _____ .

(A) $\frac{15}{64}$ (B) 1 (C) $\frac{5}{8}$ (D) $\frac{1}{16}$

13. $\frac{3}{4}\div\left(\frac{5}{8}+\frac{1}{2}\right)=$ _____ .

(A) $\frac{13}{10}$ (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) $\frac{5}{8}$

14. $\frac{3}{4}\times\left(\frac{5}{8}\div\frac{1}{2}\right)=$ _____ .

(A) $\frac{5}{8}$ (B) $\frac{2}{3}$ (C) $\frac{15}{32}$ (D) $\frac{15}{16}$

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